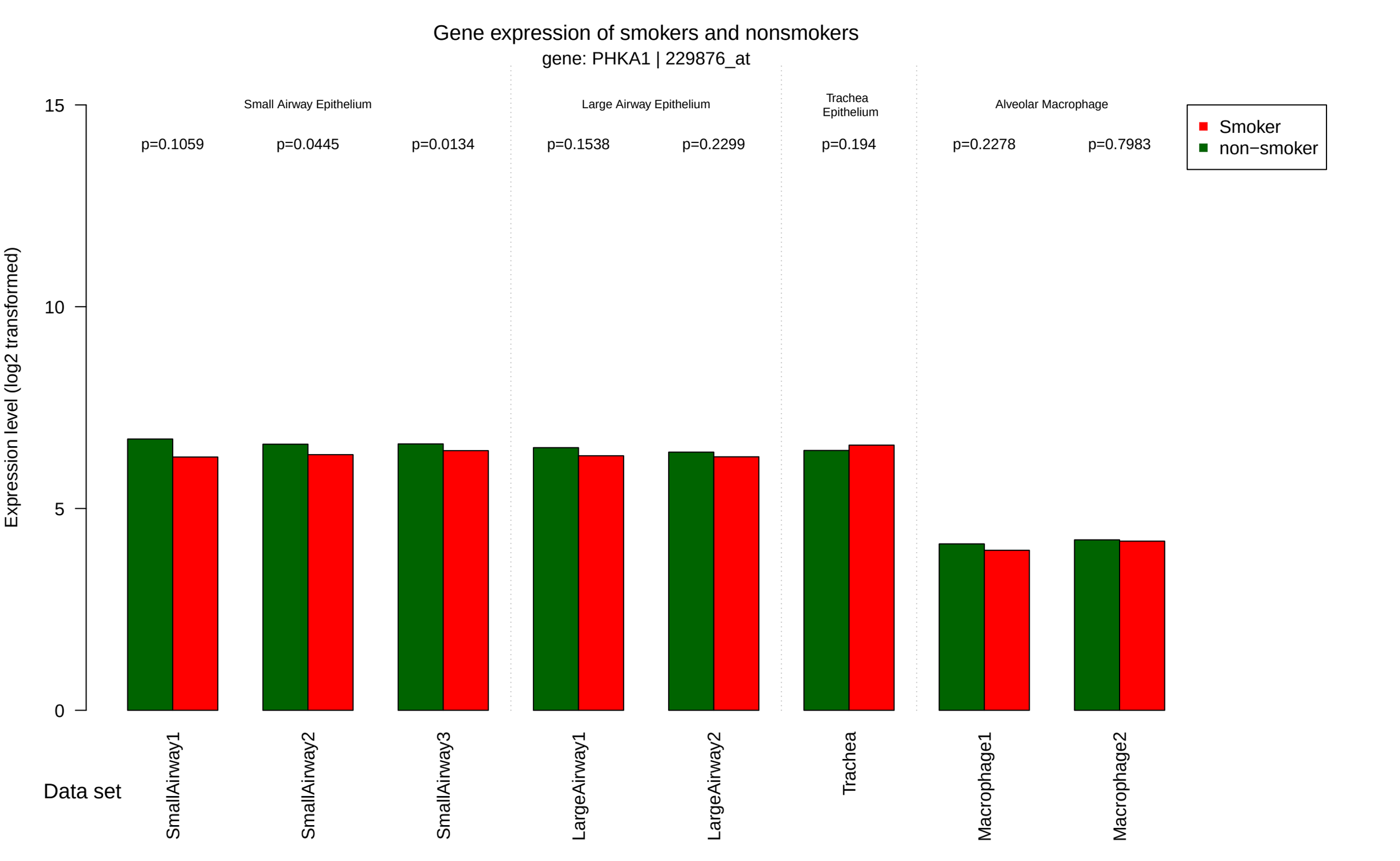
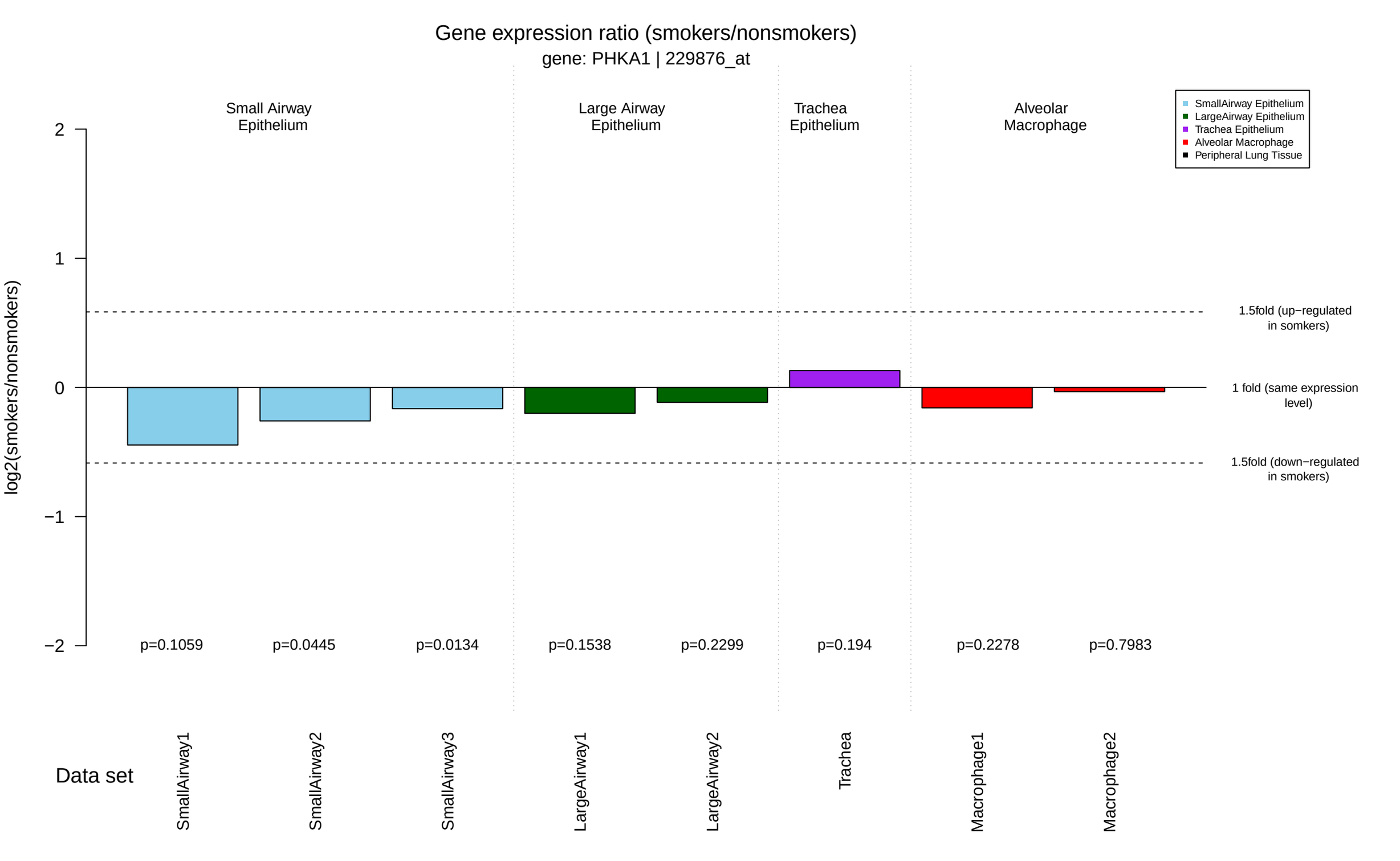
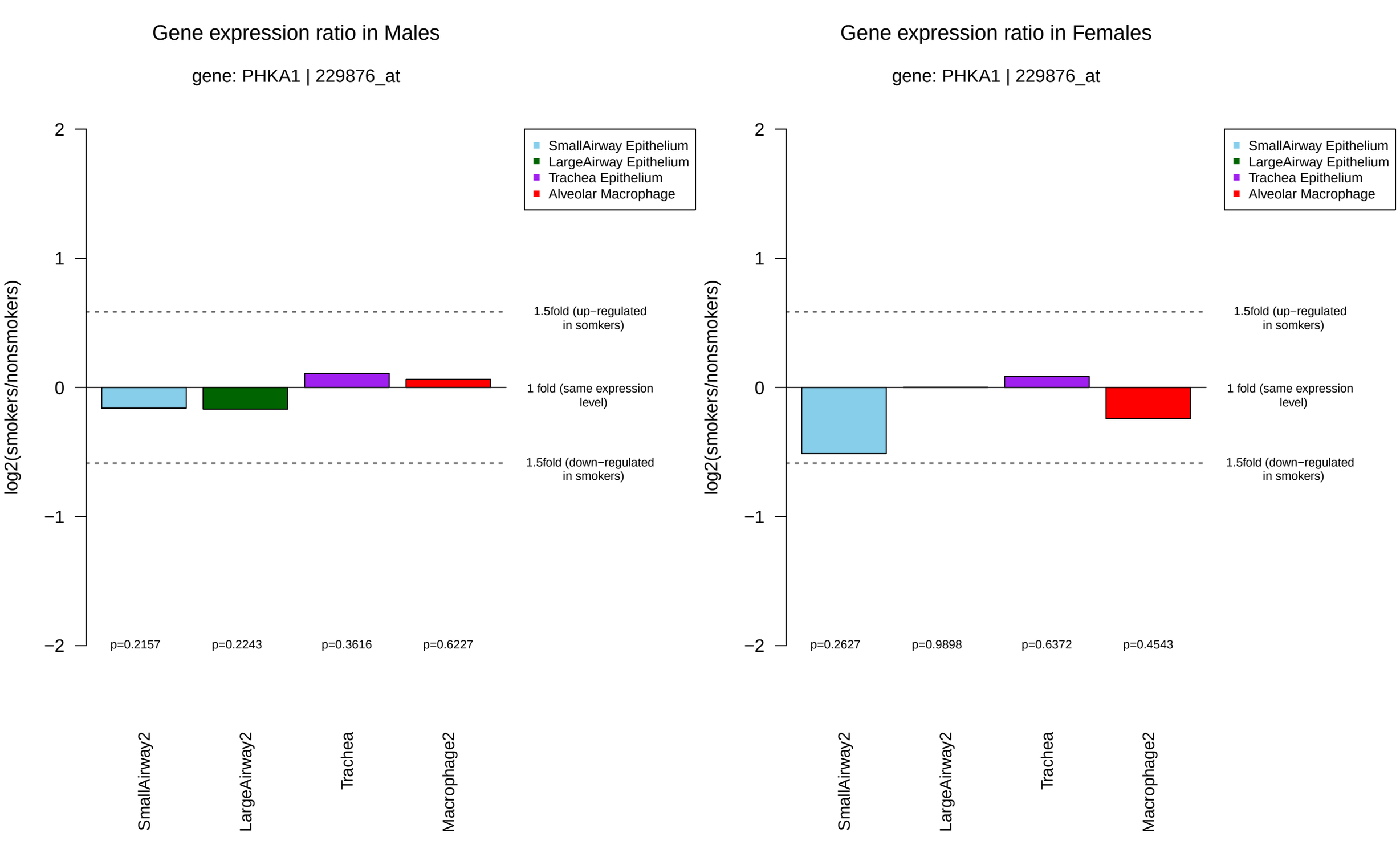
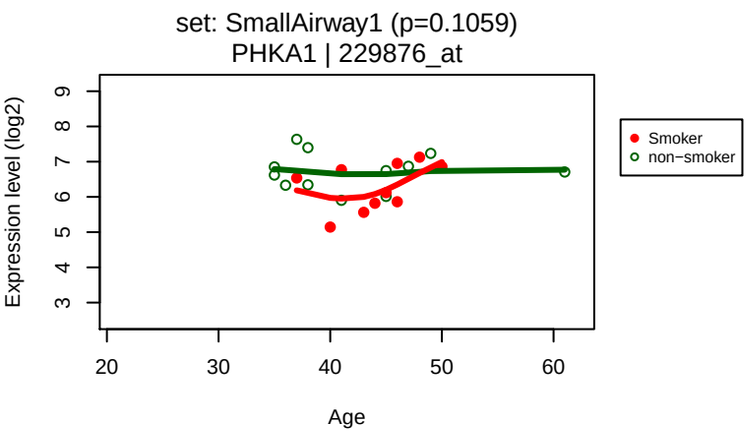
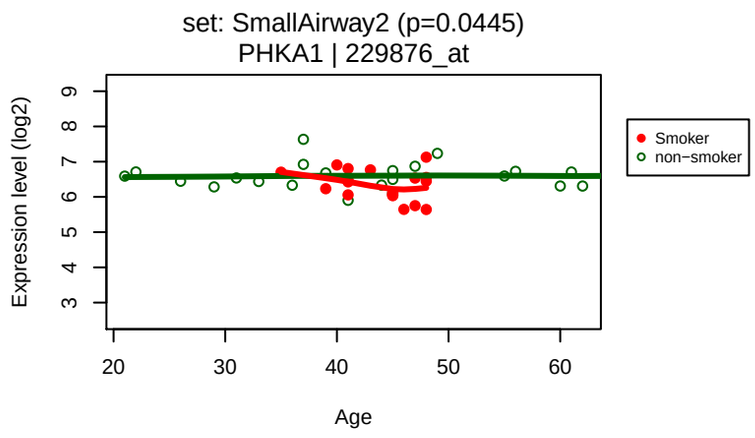
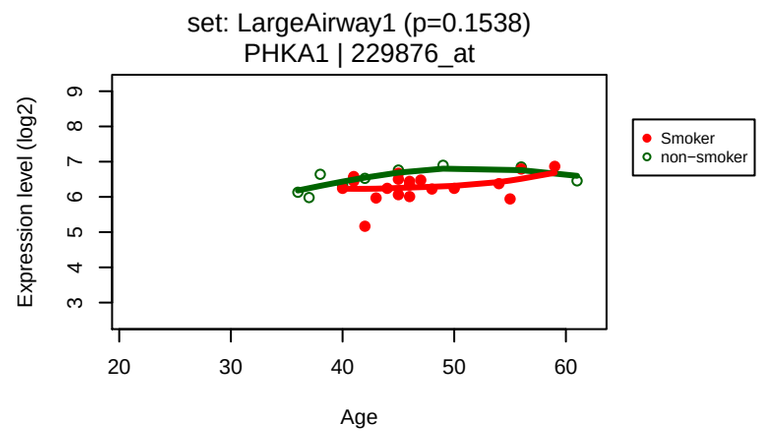

Gene expression of smokers and nonsmokers
gene: PHKA1 | 229876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1059
p=0.0445
p=0.0134
p=0.1538
p=0.2299
p=0.194
p=0.2278
p=0.7983
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

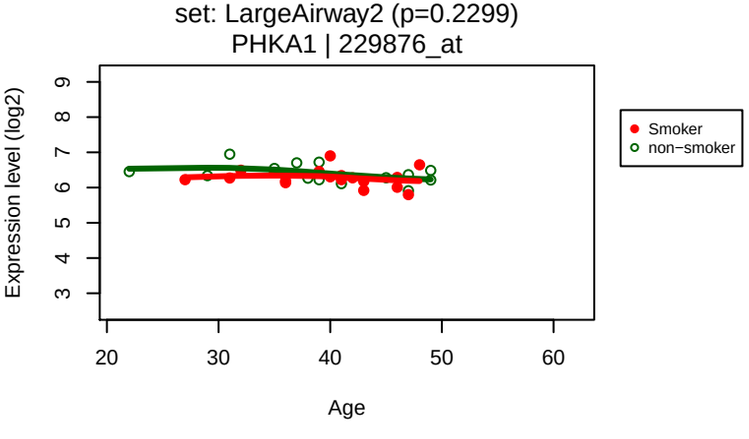

Gene expression ratio (smokers/nonsmokers)
gene: PHKA1 | 229876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1059
p=0.0445
p=0.0134
p=0.1538
p=0.2299
p=0.194
p=0.2278
p=0.7983
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

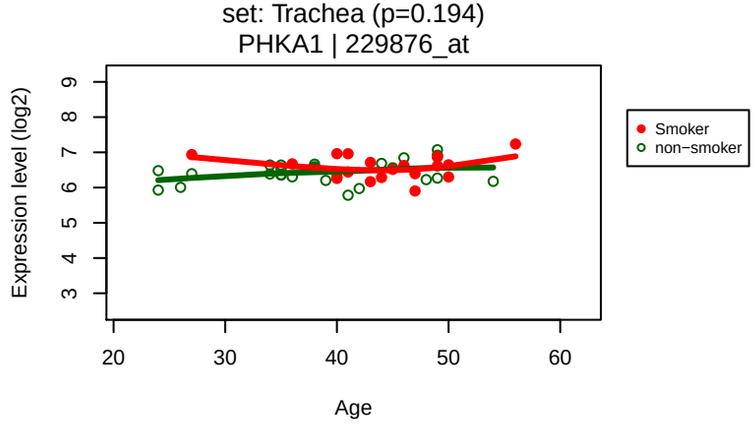

Gene expression ratio in Males
gene: PHKA1 | 229876_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2157
p=0.2243
p=0.3616
p=0.6227
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHKA1 | 229876_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2627
p=0.9898
p=0.6372
p=0.4543
SmallAirway2
LargeAirway2
Trachea
Macrophage2

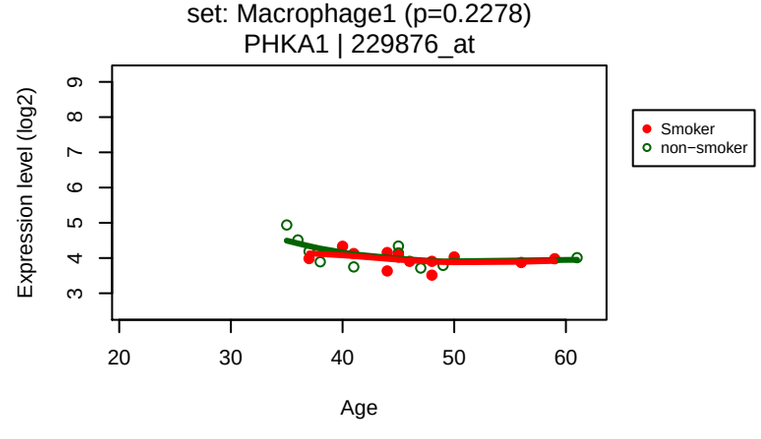

set: SmallAirway1 (p=0.1059)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker

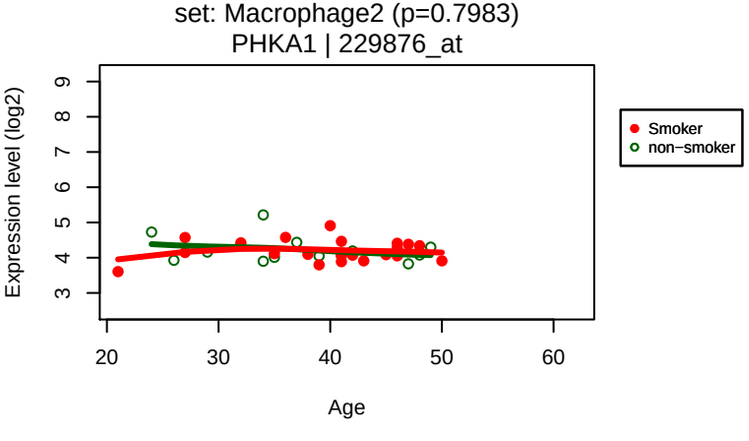

set: SmallAirway2 (p=0.0445)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker

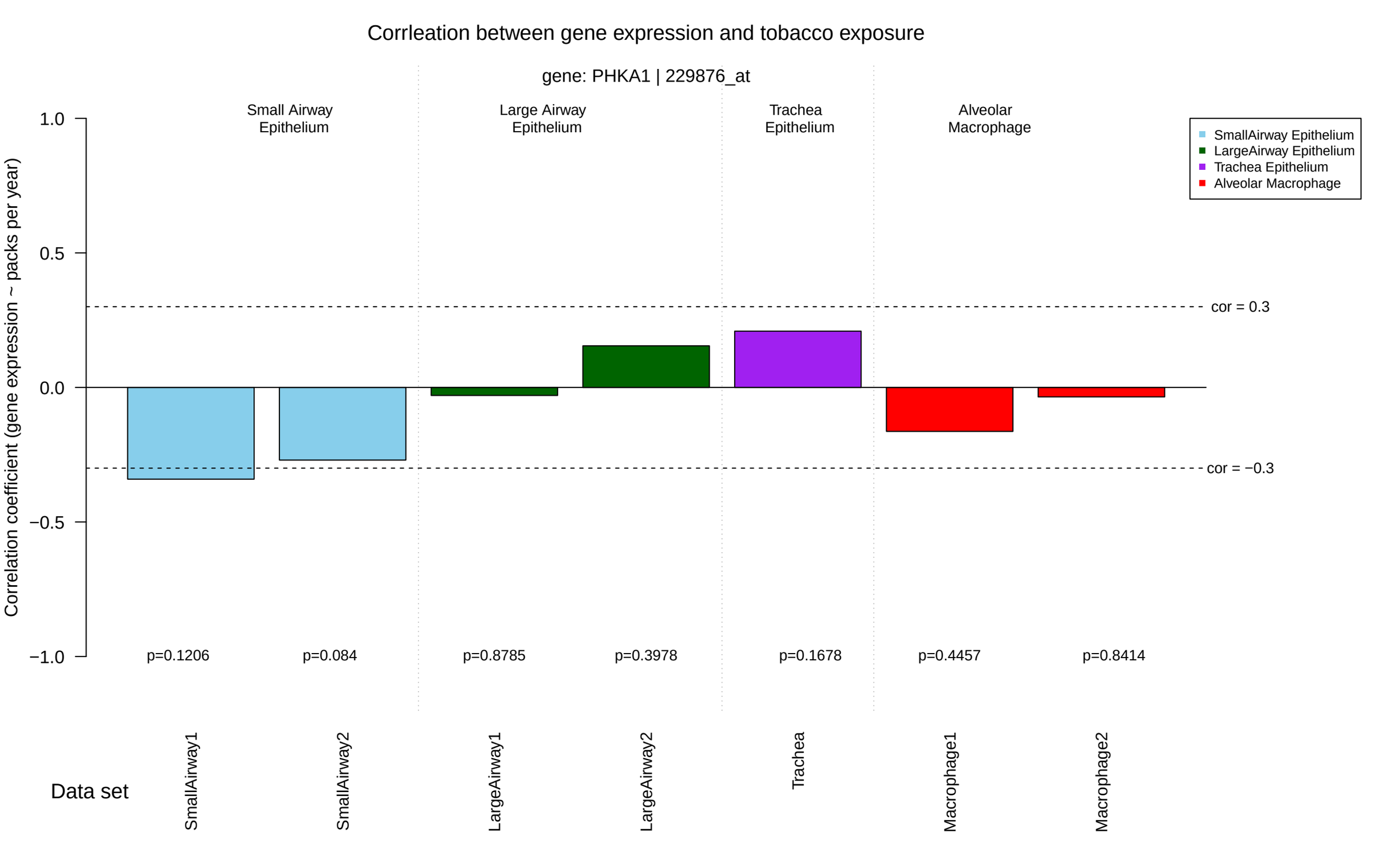

set: LargeAirway1 (p=0.1538)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2299)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.194)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2278)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7983)
PHKA1 | 229876_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHKA1 | 229876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1206
p=0.084
p=0.8785
p=0.3978
p=0.1678
p=0.4457
p=0.8414
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2